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Explainable AI-Driven Convolutional Neural Network for Predicting Electrical Isolator Defects for Smart Grid Monitoring

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***Abstract*— The reliability of electricity supply systems is greatly dependent on the health of isolators, which are responsible for safe isolation of circuits and protection of equipment. Faulty isolators can cause catastrophic consequences such as widespread blackouts, fire hazard, and serious monetary losses, hence the need for effective and proper fault detection systems. We here propose a deep learning architecture for machine isolation of electrical isolators into normal and defective classes with the assistance of transfer learning. Five pre-trained convolutional neural networks, i.e., ResNet50, DenseNet121, MobileNetV2, EfficientNetB0, and VGG16, were fine-tuned and experimented on a handpicked two-class isolator image dataset. Results indicate that all the models yielded accuracies higher than 95%, and VGG16 worked better with 97.92% accuracy and F1-scores of 98 for both classes. For increased interpretability and building up operator confidence, the use of Gradient-weighted Class Activation Mapping was made useable to give a visualization of decision-making areas. This combination of high interpretability and accuracy makes sure that the suggested method is efficient but also transparent and therefore suitable for adoption in smart grid and predictive maintenance environments.**

Keywords— Electrical Isolators, Fault Detection, Convolutional Neural Networks, Explainable AI (XAI), Power System Reliability, Smart Grid Monitoring.

I. Introduction

Electrical power infrastructure is the backbone of modern civilization, providing power to industries, telecommunication, medicine, and homes [1]. The reliability of these infrastructure systems is very much dependent on the quality of vital components, such as electrical isolators permitting to switch off the circuits to be maintained, and protection [2]. A faulty isolator may have havoc consequences, including equipment damage, mass blackout, risk for fire, huge financial losses [3]. According to recent industry reports, failure of transmission and distribution equipment is the cause of close to 25-30% of the unplanned outages globally, and one of the major causes is isolator defects [4]. In developing economies, not only is there a disruption of basic services, but also economic losses in the form of millions of dollars annually, which implies that effective detection of faulty isolators is of priority [5].

Conventional fault detection in isolators involves manual inspection, thermal imaging, or ultrasonic testing [6]. These methods are time-consuming, expensive, and prone to human mistakes, which do not scale for modern power networks [6]. With the growing prominence of smart grids and predictive maintenance, there are automated, accurate, and more and more intelligent solutions which are capable of detecting faults in real-time [7]. Artificial Intelligence (AI) and deep learning, i.e., Convolution Neural Networks (CNNs), have emerged as viable ways of doing so. Transfer learning, using pre-trained models on massive data of images, has enabled the achievement of a high accuracy in classifying faults, without too high demands on the data and computational cost.

Despite these developments, the application of artificial intelligence to power applications that have safety implications. often impeded by a lack of interpretability [8]. Although black box predictions are accurate, they do not give the transparency required by engineers and system operators to have confidence and to support model decisions [8]. To compensate for this limitation, explainable AI techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) can be utilized to highlight regions of isolator images that play a role in classification, thus Biased to offer And Actually Providing Deep Knowledge about the Decision Process.

In this work, we employ a transfer learning-based deep learning approach for automatic fault diagnosis using a carefully selected dataset of defective and healthy electrical isolators. In addition to achieving high classification accuracy, our contribution is unique in that it uses explainable AI to give a visual explanation of the predictions while ensuring that the model's focus is in line with the isolator's physical attributes. Interpretability enhances the reliability and robustness of AI-fault detection and is a step in the direction of the applied application of smart monitoring systems in the power industry.

II. Literature Review

Through efficient fault detection and isolation (FDI), electrical isolators are essential for guaranteeing power system dependability and safety. Fault detection efficacy has

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significantly increased in many domains, including distribution networks and electric drives, as sensor technologies and analysis methods have evolved over time. Techniques for fault detection and isolation are becoming more and more important for tracking, diagnosing, and fixing problems that result in expensive downtime, equipment failure, or inefficient systems.

In the field of electric drives, many researches have been undertaken on model-based approaches for the improvement of fault detection accuracy, For example, a fault detection strategy for dealing with the phase current sensors gain faults in Permanent Magnet Synchronous Motor (PMSM) drives has been proposed since such faults have the tendency to lead to disastrous operation malfunctioning and undermine mechanical device integrity [9]. Similarly robust FDI architectures on the integration of analytical redundancy relations with sliding mode observers have been promising in the aspect of solving the unexpected factory shutdown and also anticipating the degradation of quality [10]. These strategies highlight the need for balancing redundancy and robustness in order to achieve reliability for use in industrial applications.

Fault detection systems in power distribution systems have been mentioned in the literature as decentralized and intelligent. Some of the unique methodologies include Fault Detection Multi-Agent Systems (FDMAS) design, whose inception from expert logical rules facilitates the early enough detection and isolation of faults, but tolerates device failure. This allows rapid identification of faults, which minimizes service outage as well as increases the overall system reliability [11]. Such research works emphasize the scalable and adaptive FDI methods in today's power systems.

In recent years, the topic of Explainable Artificial Intelligence (XAI) has received a lot of impetus in the field of fault detection research. While deep learning and machine learning techniques achieve great accuracy on the isolator fault classification, due to their black box effects, which limit their interpretability and the difficulty of implementation in safety-critical applications [12]. Integration of XAI frameworks helps to overcome this with transparency, trust, and accountability in automatic decision-making. As stated by previous research, there is an application of XAI in the field of engineering, where it is important to know what the AI outputs are rationale for operational assurance [13]. Techniques like SHAP, LIME and Grad-CAM have been deemed useful in visualizing and exploding a model prediction. By focusing on the most important features or areas of the image, such techniques help engineers to check up on the predictions, adjust the system parameters and check that the fault detection is fairly in accordance with the domain knowledge. Thus, the inclusion of XAI in electrical isolator analysis comes with two-fold benefits: preserving high levels of predictive accuracy with the added advantage of interpretability to enlist operator confidence and enable real-world deployment [13].

The development of XAI systems promises to improve the explainability and dependability of AI-based isolator fault detection, and recent studies highlight the significance of sophisticated FDI strategies on electrical drives and power distribution networks. Our suggested method, which combines deep learning with XAI techniques to improve the accuracy and interpretability of isolator fault detection, is based on these developments.

III. Methodology

As shown in Figure 1, we employed a methodical isolation detection procedure. Initially, a dataset was gathered and is prepared for examination. To address noise, missing values, and normalization, the data were preprocessed. To guarantee an objective measurement, the dataset was split into subsets for training (80%), validation (10%), and testing (10%). Transfer Learning (TL) models were trained on the training subset, and their performance was validated and tested. The outcomes thus obtained were then analyzed to measure the stability and accuracy of the models. Finally, Explainable AI techniques were employed to provide explanations for the model decisions and enhance transparency in the detection of isolators.

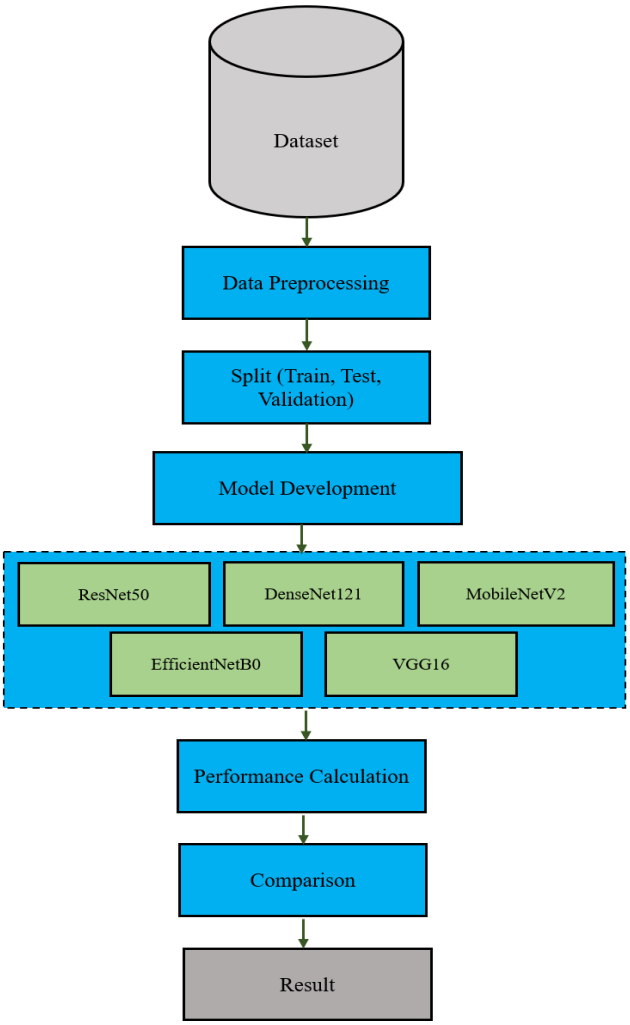


Fig. 1. Our Step-by-step Approach for Detection of Isolators Fault.

A. Dataset Description

The dataset used in this study is the inspection photographs of a 13.8 kV electric power distribution network in southern Brazil available on "https://github.com/SFStefenon/InspectionDataSet". The inspections were conducted near Blumenau city by professionals from the High Voltage Laboratory of the Regional University of Blumenau (FURB). 240 photographs were collected, of which 120 were defective structures and/or elements and 120 were healthy ones. This balanced material provides a solid foundation for the training and evaluation of deep learning algorithms for fault detection in isolators. Figure 2 shows representative samples of the dataset.

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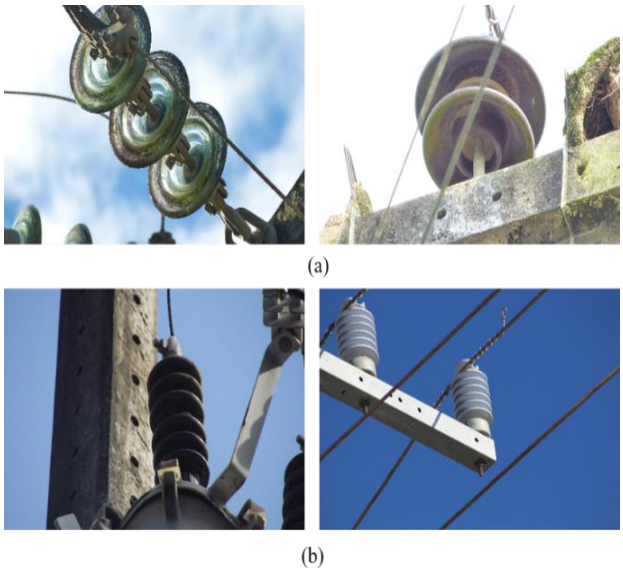


Fig. 2. Our Used Dataset Sample Images (a) Defective (b) Normal.

B. Data Preprocessing

Before the model construction, the dataset was subjected to several preprocessing steps aimed at normalizing and augmenting the dataset. The images were first downscaled to the size they would have (224 × 224 pixels), the size most transfer learning models, including VGG16, ResNet50 and EfficientNetB0, accept as input. Several data augmentation methods were then applied to enhance model generalizability and prevent overfitting because of the small size of the dataset.:

- Rotation by at most ±20° to mimic orientation variability.
- Zooming by 20% to simulate diverse inspection distances.
- Random cropping to introduce partial occlusion robustness.
- Flipping horizontally and vertically to introduce direction invariance.
- Pixel value normalization to [0,1] to allow for quick and stable convergence.

The dataset was split into 80% training, 10% validation, and 10% testing sets to provide a fair evaluation and unbiased comparison of the performances of the models.

C. Model Development

We employed five of the most commonly used transfer learning (TL) architectures:

- ResNet50: A 50-layer residual network and was modeled to address the vanishing gradient problem by skip connections (residual blocks). This means that the model is able to train a very deep architecture efficiently and ResNet50 is thus very capable of learning complex features to images due to these identity mappings.
- DenseNet121: A dense convolutional network where each layer is connected to all subsequent layers. This design improves the flow of gradients, encourages feature reuse, and has fewer parameters than conventional CNNs. DenseNet121 is particularly effective on small datasets since it is parameter-efficient and lightweight.
- MobileNetV2: A lightweight architecture that is intended for mobile and embedded vision use cases. It employs depthwise separable convolutions along with inverted residuals with linear bottlenecks, which significantly reduce computational expenses at the expense of acceptable accuracy. This is applicable in real-time defect detection under resource-constrained scenarios.
- EfficientNetB0: EfficientNetB0 was designed through neural architecture search and applies compound scaling to achieve a trade-off between depth, width, and input resolution. This makes it possible for the model to support high accuracy with fewer parameters and computational costs while being highly scalable and efficient for industrial applications.
- VGG16: A 16-layer deep convolutional neural network characterized by its simplicity and uniform use of 3×3 convolution filters. Despite being an earlier architecture, VGG16 remains highly effective in transfer learning tasks due to its ability to capture hierarchical visual features. In our study, VGG16 achieved the best overall performance among the tested models.

D. Explainable AI (GradCAM)

While high accuracy is important, model interpretability is critical for building trust in real-world applications. To achieve this, we applied Grad-CAM (Gradient-weighted Class Activation Mapping) on the best-performing model, VGG16. Grad-CAM produces visual heatmaps that point towards the parts of an image that are most important in the decision-making process by a model. This helped us ensure that the model predictions were done based on corresponding defective areas of the isolators and not on noise in the background. The incorporation of Explainable AI not only improved model explainability but also improved the reliability of the proposed defect detection system.

E. Performance Calculation

We evaluated performance using various metrics to identify the most effective classifier for detecting isolator defective. Performance indicators, expressed as percentages (%), were calculated using Eqs. (1–4). Additionally, we generated a confusion matrix for each model to assess their performance comprehensively

$$Accuracy = \frac{True\ Positive + True\ Negative}{All\ Instance} \quad (1)$$

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad (2)$$

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \quad (3)$$

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

IV. Analysis and Discussion

In this study, we employed transfer learning using five pre-trained convolutional neural network models—ResNet50, DenseNet121, MobileNetV2, EfficientNetB0, and VGG16—

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trained on 10 epochs for the isolator dataset. For comparison of performance, we compared primary classification metrics like precision, recall, F1-score, and overall accuracy, along with confusion matrices and receiver operating characteristic (ROC) curves with area under the curve (AUC) values. Finally, the best-performing model was again analyzed using explainable AI techniques, Gradient-weighted Class Activation Mapping (Grad-CAM), to determine the decision-making process.

Comparison of the performance of five models of transfer learning—ResNet50, DenseNet121, MobileNetV2, EfficientNetB0, and VGG16—demonstrates that each architecture achieves better performance in classifying electrical isolators into defective and normal categories. Based on Table 1, the models have always achieved accuracy greater than 95%, with balanced precision, recall, and F1-score for both classes. ResNet50, DenseNet121, MobileNetV2, and EfficientNetB0 produced extremely close results, scoring around 95.8% accuracy, with perfect precision in the defective class and perfect recall in the normal class. VGG16 had the best performance, achieving the highest accuracy of 97.92% and F1-scores of 98 for both classes. Such improved balance is indicative of the ability of VGG16's convolutional hierarchy to learn more discriminative features from isolator images and thereby reduce misclassifications as well as enjoy superior generalization.

TABLE I. PERFORMANCE OF APPLIED TRANSFER LEARNING MODEL FOR ISOLATORS DEFECTIVE DETECTION.

Model Name	Class	Precision	Recall	F1-Score	Accuracy
ResNet50	Defective	100	92	96	95.82%
	Normal	92	100	96	
DenseNet121	Defective	100	92	96	95.85%
	Normal	92	100	96	
MobileNetV2	Defective	100	92	96	95.84%
	Normal	92	100	96	
EfficientNetB0	Defective	100	92	96	95.83%
	Normal	92	100	96	
VGG16	Defective	100	96	98	97.92%
	Normal	96	100	98	

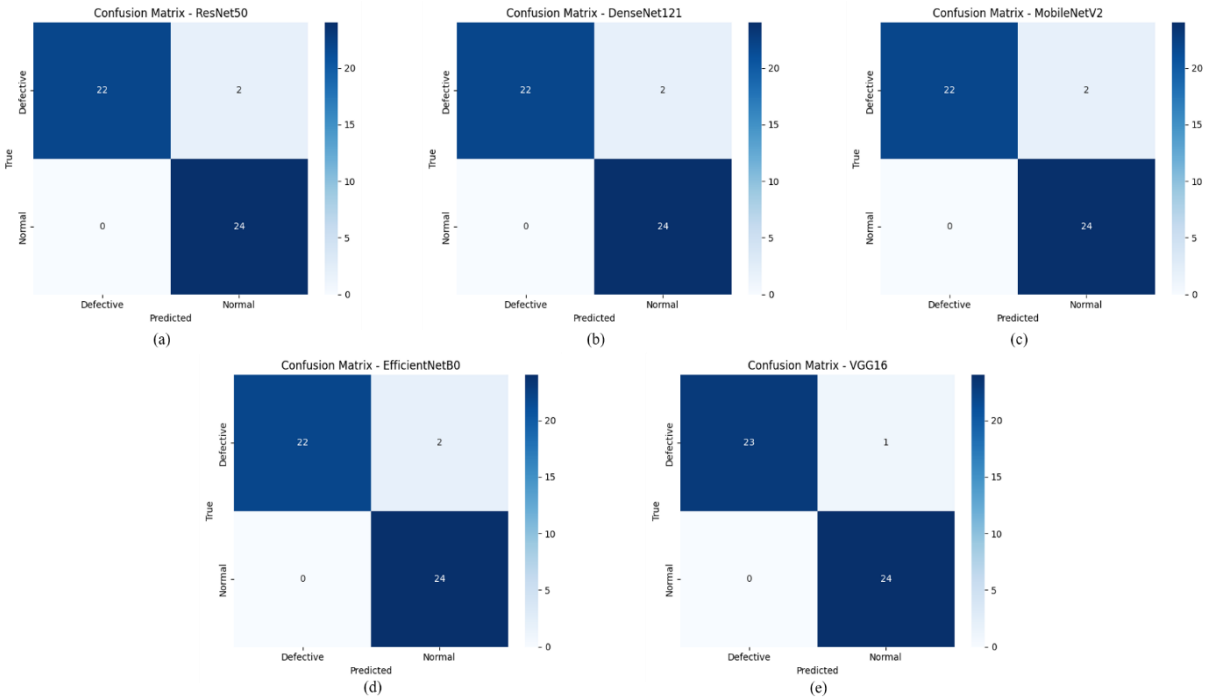


Fig. 3. Confusion Matrix of Applied Models for Isolators Defective Detection (a) ResNet50 (b) DenseNet121 (c) MobileNetV2 (d) EfficientB0 (e) VGG16.

The Figure 3 confusion matrices provide a more detailed image of these results by specifying the division of correct and incorrect predictions of the two classes. Two faulty isolators were mistakenly marked as normal for ResNet50, DenseNet121, MobileNetV2, and EfficientNetB0, while all normal isolators were detected accurately. VGG16, conversely, kept misclassification to only one faulty sample, with perfect identification of the normal class. This small but substantial decrease in false negatives is crucial in practical applications because missing a faulty isolator is more dangerous to power system reliability than a false alarm. As a result, VGG16 shows a greater capacity to reduce critical errors, further demonstrating its applicability in real-world monitoring systems.

The examination of the ROC curve in Figure 4 provides additional confirmation of the models' discriminative power. The models' stability and dependability in differentiating between faulty and normal isolators are confirmed by their AUC values, which were all around 1.00 or 0.97. With the highest AUC of 1.00, DenseNet121 demonstrated superior classification performance across a range of decision thresholds. ResNet50 and EfficientNetB0 trailed closely behind with AUCs of 0.99 and 0.98, respectively. While VGG16 achieved the highest accuracy and F1-scores in earlier

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tests, its AUC of 0.97 suggests comparatively lower stability than DenseNet121. However, ROC analysis overall verifies that all five models exhibit high generalization capability with minimal loss in sensitivity over specificity.

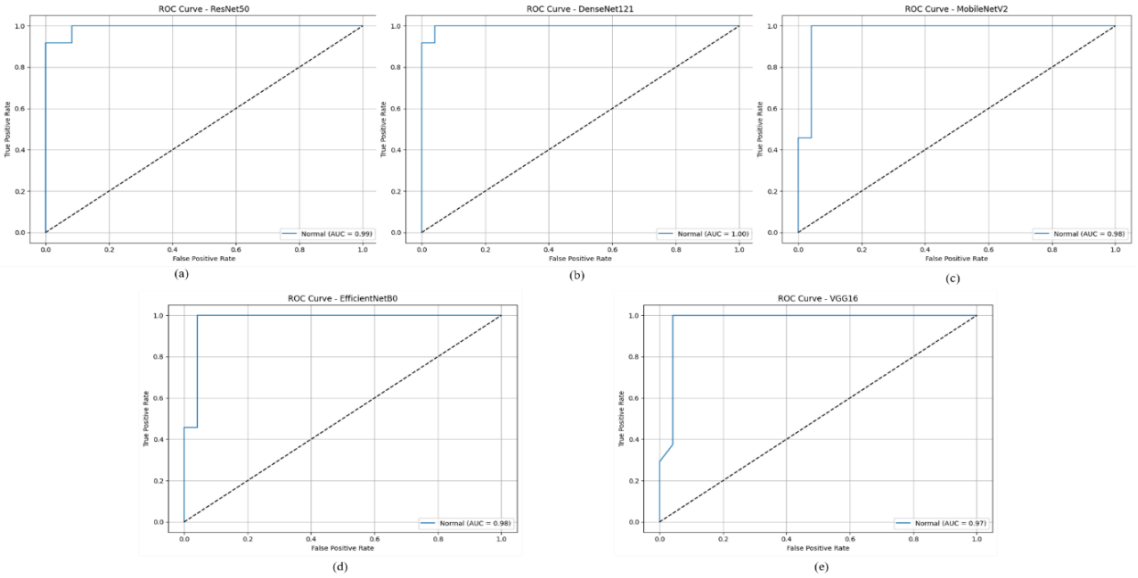


Fig. 4. ROC Curve of Applied Models for Isolators Defective Detection (a) ResNet50 (b) DenseNet121 (c) MobileNetV2 (d) EfficientB0 (e) VGG16.

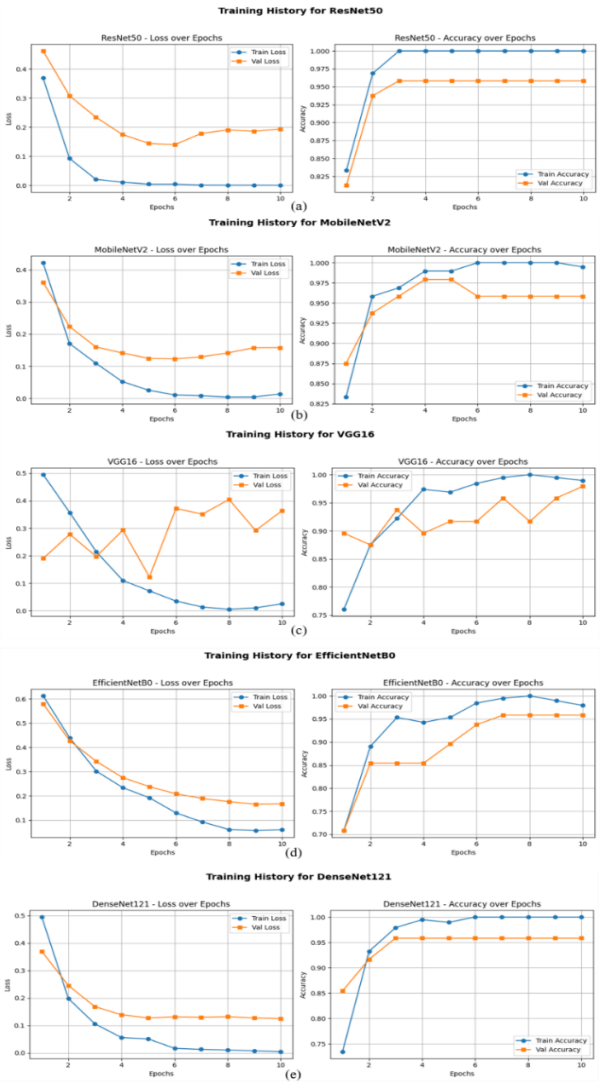


Fig. 5. Training History of Applied Models for Isolators Defective Detection (a) ResNet50 (b) MobileNetV2 (c) VGG16 (d) EfficientB0 (e) DenseNet121.

The training curves of models in Figure 5 show how the models train over 10 epochs. For ResNet50, DenseNet121, MobileNetV2, and EfficientNetB0, both training loss and validation loss reduced gradually, while accuracies increased sharply and plateaued after a few epochs, reflecting efficient convergence and effective transfer of pre-trained weights from ImageNet features. For VGG16, validation loss and accuracy show greater fluctuations, reflecting data variation sensitivity. Despite these fluctuations, VGG16 ultimately had the highest validation accuracy among all models, vindicating its superior classification.

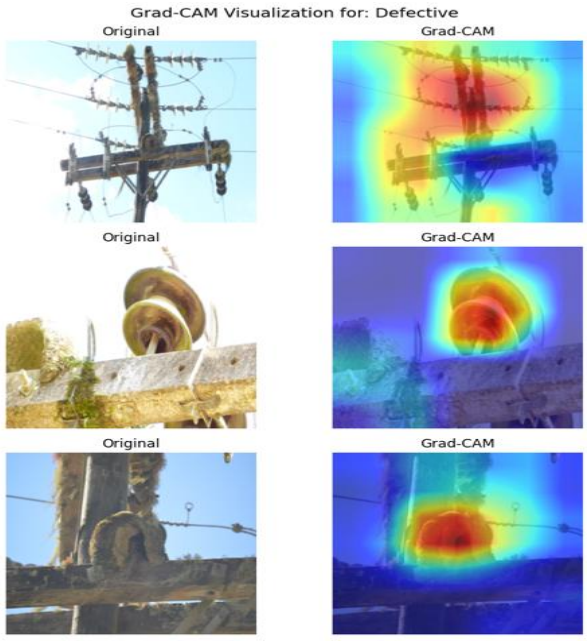


Fig. 6. Explainable AI (Grad CAM) with VGG16 for Interpretability on Defective Isolators.

For greater interpretability, Grad-CAM was employed to overlay the decision-making regions of the top-performing model, VGG16, for faulty isolators (Figure 6). The heatmaps

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clearly highlight areas of structural anomalies such as cracks, surface contamination, and mechanical stress, thereby confirming that the predictions of the model were being made on physically relevant features. For instance, insulator discs with observable eye dirt deposits or damaged connectors were strongly weighted in the activation maps, attesting that the network was able to successfully localize the fault-related regions.

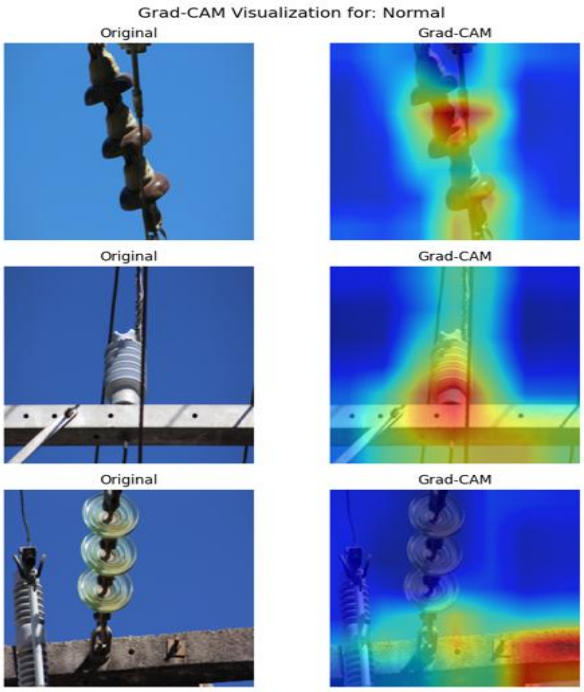


Fig. 7. Explainable AI (Grad CAM) with VGG16 for Interpretability on Normal Isolators.

Grad-CAM on normal isolators (Figure 7) indicates that the attention of the model was distributed across intact bodies, discs, and connectors without the presence of localized hotspots characteristic of damage. The heatmaps reflect smooth and uniform activation patterns, as expected by the structural integrity and continuity of healthy isolators. The consistent focus on the defect-free surfaces confirms that the model learns to project unperturbed structural features onto the normal class, such that its prediction is unaffected by the irrelevant background regions.

V. Conclusion

The proposed research in this study indicates a transfer learning-based method for electrical isolator classification as normal or faulty to fulfill the requirement of automatic fault detection in power systems. Five pre-trained models (ResNet50, DenseNet121, MobileNetV2, EfficientNetB0, and VGG16) all achieved over 95% accuracy, with VGG16 performing best at 97.92% and having a good balance between precision, recall, and F1-score. Grad-CAM visualizations set the focus of the model on deteriorated regions for defective isolators and healthy regions for healthy ones, corresponding to human diagnosis, and making it more reliable. The system is easy to implement in smart grids with minimal manual inspection, enhanced safety, and reduced economic losses. Larger datasets, real-time implementation, and multi-class fault identification will be the focus of future work for large-scale industrial use.

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